Dhruv Kithany

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Education

University of Wisconsin–Madison Madison, WI

B.S. Computer Science, B.S. Statistics *Aug 2025 – May 2028*

- **Minors/Certificates:** Mathematics, Data Science
- **Relevant Coursework:** Data Structures & Algorithms, Intro to Artificial Intelligence, Data Management for Data Science, Data Science Modeling I & II, Theory of Probability, Discrete Mathematics, Multivariable Calculus & Linear Algebra, R for Statistics I & II, Computer Architecture & Computer Engineering, CMU 15-122, CMU 15-112

Eagle Scout | *Boy Scouts of America; led **165-hour** wildlife shelter project* *May 2020 – Feb 2025*

Experience

Software Developer Intern, Enterprise Architecture May 2026 – Aug 2026

Federated Hermes *Pittsburgh, PA*

- Architected and built **SIRIS**, a company-wide production multi-agent Retrieval-Augmented Generation assistant (Copilot Studio, Azure OpenAI/GPT-5.5, Microsoft Entra ID, SharePoint, Dataverse) with a router directing to **6 independently-grounded specialist agents**; identity-scoped retrieval added **zero net-new access permissions**
- Authored **165+ rubric-based evaluation cases** across 5 suites, verifying all 6 ARB specialists and **20/20 passing test cases** on a Compliance proof-of-concept that generalized the architecture to a second department; resolved **5+ platform defects**, including re-architecting a 3-level design into a 2-level hub after hitting a connected-agent chaining ceiling
- Built **OneView**, a Power BI report consolidating **2,500+ governance artifacts** across 6 categories onto one page via a semantic model over live Databricks HRView data, a Power Automate/Graph identity-sync pipeline, and SharePoint

Undergraduate ML Researcher Sept 2026 – Present

Auditory Encoding and Processing Optimization (AEPO) Lab, Waisman Center (PI: Dr. Bobby Gibbs) *Madison, WI*

- Built a **dual-pipeline speech-masking engine** in the lab's BubbleMixer library, adding LibriSpeech speech-in-speech masking (polyphase resampling, raised-cosine ramps) alongside bubble noise, with a PyTorch `Dataset`-compatible worker
- Parallelized synthesis with `asyncio` worker threads and an 8-task lookahead generator, raising throughput from **1.5 to 12 files/sec** (1,000 mixtures in **1.4 min vs. 11 min**) while capping in-flight memory at ∼**17 MB**
- Added validation to WavLM training on live-synthesized audio with an **80/20 phoneme-stratified split** to prevent leakage, plus epoch-level val loss/accuracy and per-step loss logging
- Fit R mixed-effects models and built a **10-figure deck** for the PI on **16 subjects**, showing broad cochlear-implant current spread needs ∼**3.3x more bubbles/sec** and cuts sensitivity (d-prime) by **13.6%**

Undergraduate ML Researcher Sept 2026 – Present

Integrated Diagnostics and Analytics (IDiA) Lab for Precision Medicine (PI: Dr. Pallavi Tiwari) *Madison, WI*

- Developing deep-learning pipelines for MRI stroke-lesion segmentation, benchmarking **nnU-Net v2, SwinUNETR, and prompted SAM** on open datasets (SOOP, ISLES 2022) with frozen patient-level splits and a locked UW clinical holdout
- Setting up reproducible Docker-based training on UW's CHTC cluster with standardized evaluation (Dice, lesion-wise F1, HD95)

Research Software Engineer June 2026 – Oct 2026

Farm 2 Facts (Advisor: Dr. Morales) *Madison, WI*

- Developed backend infrastructure for an open-source Ruby web application supporting **100+ specialty crop farmers across 5 states** in a multi-institutional NSF/USDA consortium, enabling streamlined USDA grant reporting
- Integrated REST APIs for Wisconsin Mesonet weather data; prototyped natural-language querying and dashboards

Undergraduate AI Researcher, LLM Evaluation Nov 2025 – Present

Wisconsin Undergraduate Research in Data Science (WISCURDS), Advisor: Dr. Adam Ross Nelson *Madison, WI*

- Data manager and principal architect of all data flows for an LLM synthetic-human-proxy study; built Python pipelines for automated cleaning, NaN auditing, lineage receipts, and versioned inventory across **66+ datasets** and **133,000+ data cells**
- Authored all **11** reproducible Quarto research memos (ANOVA, EFA, Wasserstein distance) on LLM survey-response bias
- Built a from-scratch JavaScript factor analysis engine (Jacobi, parallel analysis, Promax) and dashboard across **25+ language models**

Shadow Software Developer May 2023 – Aug 2025

Federated Hermes *Pittsburgh, PA*

- Built Python FX trading simulators and LSTM equity-forecasting prototypes (TensorFlow/Keras) for portfolio management; researched blockchain asset management over three consecutive summers

Projects

EEG-Based Emotion Decoding Pipeline – (MindMesh) | *Python, PyTorch, Scikit-learn, Signal Processing*

- Built an end-to-end machine learning pipeline decoding **27 fine-grained emotions** from EEG signals across **88 subjects and 11,000+ trials**, reaching **30.83% Top-1 accuracy (8x random chance)** with zero cross-subject data leakage
- Benchmarked 6 classifier and neural-network architectures, reaching **80%+ Top-5 accuracy** across 27 emotion classes with **8.8ms inference latency** for the top-performing model

Real-Time Presentation Coaching Tool – (Lumina-Presenter) | *Python, PyQt6, MediaPipe, OpenCV, WebRTC*

- Built a real-time computer vision pipeline tracking a 468-point 3D face mesh and 33 skeletal landmarks to detect gaze drift and fidgeting, surfaced through a transparent always-on-top HUD overlay
- Added a speech layer with WebRTC voice-activity detection and speech recognition to sync script progression and trigger an LLM-powered (Groq API) Q&A probe system during detected pauses

C0 Bytecode VM & Execution Visualizer – (StackScope) | *C, Python, JavaScript* Live Demo

- Built a stack-based bytecode VM in C implementing **45 opcodes** of the C0 ISA with call frames, bounds-checked heap arrays, and **12 classes of runtime traps** via `setjmp`/`longjmp`; sustains ∼**525 MIPS** (median of 15 runs)
- Validated by **48 automated tests** at **96% line coverage** of the VM core, with CI on Ubuntu and Windows
- Wrote a Python C0-subset compiler and disassembler, plus a zero-dependency time-travel web visualizer showing the operand stack, call frames, and heap with explanations for all **45 opcodes**

Technical Skills

Languages: Python, Java, C/C++, SQL, NoSQL, JavaScript/TypeScript, R, Ruby, HTML/CSS

Frameworks & Libraries: TensorFlow, Keras, PyTorch, Scikit-learn, Pandas, NumPy, OpenCV, MediaPipe, React, Node.js

Cloud & Data: Azure, Azure OpenAI, Databricks, Power BI, Power Automate, MongoDB, MySQL

Dev Tools: Git, GitHub Actions, pytest, REST APIs, Microsoft Graph API, Copilot Studio